

Nassim TAZIR

Data Scientist

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PROFILE

Data Scientist with a background in Applied Data Science (Collège La Cité) and a Master's degree in Forecasting and Economic Analysis. Experienced in developing predictive models, feature engineering, and rigorous model validation on complex datasets. Strong interest in artificial intelligence and large language models (LLMs), with active monitoring of generative AI trends. Proven ability to communicate and explain technical concepts to non-technical audiences.

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, scikit-learn, Matplotlib, Seaborn, PySpark), R, SQL

Machine Learning: Random Forest, XGBoost, LightGBM, K-Means, regression, classification, feature engineering, sklearn pipeline, Optuna, SHAP

Time Series: TimeSeriesSplit, temporal cross-validation, cyclical encoding, lags, rolling features, custom metrics (WMAE)

Big Data / ETL: Hadoop MapReduce, HDFS, PySpark, Streamlit, Docker, MySQL, SQL Server

BI / Visualization: Power BI, Tableau, Excel

Project Management: Agile (Scrum, Kanban), Lean Six Sigma Green Belt, PMP

ACADEMIC PROJECTS - COLLÈGE LA CITÉ

Weekly Sales Forecasting System - Walmart

Tools: Python, XGBoost, LightGBM, Optuna, SHAP, scikit-learn, pandas, TimeSeriesSplit

- End-to-end forecasting pipeline on 408,000+ transactions across 45 stores and 81 departments (2010-2012).
- Advanced feature engineering: sin/cos cyclical encoding, lag features (Lag_1, Lag_2, Lag_4), rolling statistics, dedicated holiday flags (Black_Friday, Thanksgiving, Xmas_Week), Target Encoding without data leakage.
- Comparison of 5 models using a custom WMAE metric (x5 weight on holiday weeks). WMAE improved from \$4,249 to \$1,591 (-62.6%). Final model: XGBoost, R2: 0.97.
- Variable selection via SHAP and validation using strict temporal split (train 2010-2011, test 2012).

[GitHub](#) [Portfolio](#) [Presentation Video](#)

Credit Card Default Risk - Segmentation and Analysis (UCI)

Tools: Python, pandas, scikit-learn, Prince (PCA/CA/MCA), KMeans, Seaborn, Matplotlib

- Exploratory analysis on 30,000 clients, PCA (90% of variance explained), CA and MCA for categorical variables.
- K-Means segmentation (k=4), identification of a cluster with 62.6% default rate. Comparison of 3 approaches via radar chart.

[GitHub](#) [Portfolio](#)

House Price Regression Model - R

Tools: R, leaps, MASS, car, lmtest

- Multiple linear regression on 1,728 observations, stepwise variable selection (leaps), residual diagnostics and assumption validation (heteroscedasticity, multicollinearity). Adj. R2: 0.583.

[GitHub](#) [Portfolio](#) [Presentation Video](#)

Big Data Pipeline - Sales Analysis with Hadoop MapReduce and Streamlit

Tools: Python, Hadoop (HDFS, YARN), MapReduce, Streamlit, Docker, Docker Compose

- Dockerized Hadoop cluster (1 master + 2 workers), 4 MapReduce jobs with HDFS storage and interactive Streamlit dashboard.

[GitHub](#) [Portfolio](#)

WORK HISTORY

Customer Experience Associate - ScotiaBank - Apr. 2026 - Present

- Retail banking environment in the financial services sector, with exposure to investment products (TFSA, RRSP, FHSA) and financial regulations (FINTRAC, PIPEDA).

Data Analyst - Financial Services - PROIMP (Analytical Services) - Apr. 2020 - Feb. 2024

- Development of predictive models and segmentation on a vehicle financing portfolio (~3,700 files) using Python and R.
- K-Means clustering to identify high-risk groups, tracking of key KPIs and data quality control via SQL.
- Production of automated dashboards and reports to support portfolio decisions.

Other activity (Oct. 2021 - Sept. 2024): Delivery of project management training and coaching for PMP and Scrum certifications.

Automotive Network Development Manager - KIA Motors, Algeria - Aug. 2017 - Mar. 2020

- Data collection, structuring and analysis of the vehicle fleet to identify network expansion opportunities. Preparation of reports and dashboards for senior management.

Distribution Network Development Officer - RENAULT, Algeria - Dec. 2012 - Feb. 2016

- Data collection and quality control for partner infrastructure tracking. Preparation of reports and project follow-up.

EDUCATION

Diploma of Collegial Studies - Applied Data Science - Collège La Cité, Ottawa - Sept. 2024 - Apr. 2026

Master's Degree in Forecasting and Economic Analysis - Institut Supérieur de Gestion et de Planification, Algeria - 2021 - WES Equivalency Ref. 5666449IMM

CERTIFICATIONS

PSM I - Professional Scrum Master - PSPO I - Professional Scrum Product Owner - LSSGB - Lean Six Sigma Green Belt - PMP - Project Management Professional - CESAM - Applied Statistics (Feb. 2024)

INTERPERSONAL SKILLS

Bilingual French and English - Communication and simplification of technical concepts - Collaborative teamwork in multidisciplinary teams - Autonomy and analytical rigor - Intellectual curiosity and continuous learning